

Ahmed S. Elwakil

professor

基本信息

姓名: Ahmed S. Elwakil

所属院校: 卡尔加里大学

地区: Canada

海外经历: 有

学科领域

集成电路设计与集成系统

电气工程及其自动化

生物医学工程

电子信息工程

微电子科学与工程

研究领域

模拟与射频IC设计

EDA工具与电路验证

电力系统仿真与优化

分布式发电与储能技术

电子电路与系统

生物信号处理与模式识别

模拟与数模混合电路

混合信号集成技术

信号与信息处理

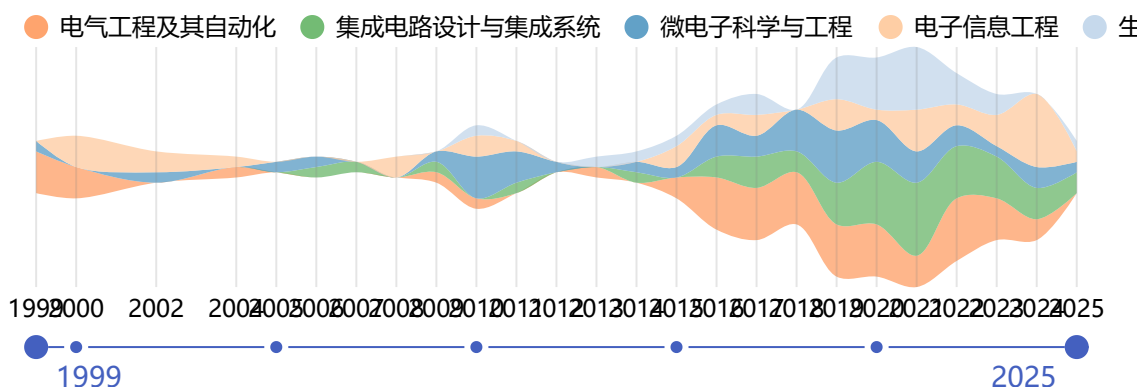
电力电子与电机驱动

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个人简介

Ahmed S. Elwakil is a professor of Electrical Engineering at the University of Sharjah, the University of Calgary, and Nile University. His research areas include Circuit Theory, Nonlinear Circuits, Fractional - Order Systems, Bioimpedance, and Super - capacitors. He has an h - index of 54 (35 since 2020) and an i10 - index of 228 (142 since 2020). He has co - authored numerous papers, such as "Fractional - order circuits and systems: An emerging interdisciplinary research area" with high citation numbers. His email has been verified at sharjah.ac.ae.

近年发表的论文



Closed-Form Expressions for the Input Impedance of Some 2-D Fractal Circuit Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Elwakil, Ahmed S.; Allagui, Anis; Elamien, Mohamed B.; Psychalinos, Costas; Maundy, Brent

被引用次数: 0

2025

Third-order resonance networks and their application to chaos generation

INTEGRATION-THE VLSI JOURNAL

Elwakil, Ahmed S.; Maundy, Brent J.; Psychalinos, Costas; Elsonbaty, Amr

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2025

Non-integer-order exponential filters: definitions in the time-domain and frequency-domain

NONLINEAR DYNAMICS

Nako, Julia; Psychalinos, Costas; Elwakil, Ahmed S.

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Analog filters based on the Mittag-Leffler functions

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Review and novel contributions to amplifier-based oscillator design

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Maundy, Brent J.; Elwakil, Ahmed S.; Elamien, Mohamed B.; Psychalinos, Costas

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A novel family of tunable-frequency oscillators

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Systematic design of negative group delay circuits

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2024

Reduced Complexity Approximation and Design of Gaussian Impulse Response Filters and Wavelets

IEEE ACCESS

Nako, Julia; Tsimokou, Georgia; Psychalinos, Costas; Elwakil, Ahmed S.; Maundy, Brent J.; Bertsias, Panagiotis

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Bilinear Double-Order Filter Designs and Application Examples

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Nako, Julia; Psychalinos, Costas; Khateb, Fabian; Elwakil, Ahmed S.

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Procedure for Obtaining the Analytical Distribution Function of Relaxation Times for the Analysis of Impedance Spectra Using the Fox H-Function

JOURNAL OF PHYSICAL CHEMISTRY C

Allagui, Anis; Elwakil, Ahmed S.

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2024

On the Theory and Application of the Fractional-Order Dirac-Delta Function

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Allagui, Anis; Elwakil, Ahmed S.

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Chaos in Inter-State-Controlled RLC Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Elwakil, Ahmed S.; Allagui, Anis; El-Mesady, Ahmed; Elsonbaty, Amr; Majzoub, Sohaib; Maundy, Brent J.

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A note on the bandwidth of negative group delay filters

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On the Behavior of a Non-Linear Bandpass Filter with Self Voltage-Controlled Resistors

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Fractional-Order Equivalent-Circuit Model Identification of Commercial Lithium-Ion Batteries	
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Non-Integer Order Generalized Filters Designs

IEEE ACCESS

Nako, Julia; Psychalinos, Costas; Elwakil, Ahmed S.; Minaei, Shahram

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Design of Higher-Order Fractional Filters With Fully Controllable Frequency Characteristics

IEEE ACCESS

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2023

Time-domain response of supercapacitors using their impedance parameters and Fourier series decomposition of the excitation signal

JOURNAL OF ELECTROANALYTICAL CHEMISTRY

Allagui, Anis; Fouda, Mohammed E.; Elwakil, Ahmed S.; Psychalinos, Costas

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MICROELECTRONICS JOURNAL

Bertsias, Panagiotis; Psychalinos, Costas; Minaei, Shahram; Yesil, Abdullah; Elwakil, Ahmed S.

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2023

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INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

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2023

Field Programmable Analog Array Based Non-Integer Filter Designs

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2023

Electrical Impedance Spectroscopy Using a Wide-Band Signal Based on the Rudin-Shapiro Polynomials

JOURNAL OF THE ELECTROCHEMICAL SOCIETY

Al-Ali, Abdulwadood; Elwakil, Ahmed S.; Maundy, Brent; Majzoub, Sohaib; Allagui, Anis

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Wide Bandwidth Signals for Joint Time–Frequency Characterization of Nonlinear and Time-Varying Circuits

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2023

Minimum Active Component Count Design of a PI λ D μ Controller and Its Application in a Cardiac Pacemaker System

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Electronically Controlled Power-Law Filters Realizations

FRACTAL AND FRACTIONAL

Tsouvalas, Errikos; Kapoulea, Stavroula; Psychalinos, Costas; Elwakil, Ahmed S.; Jurisic, Drazen

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Numerical Sensitivity Analysis and Hardware Verification of a Transiently-Chaotic Attractor

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Sayed, Wafaa S.; Mohamed, Sara M.; Elwakil, Ahmed S.; Said, Lobna A.; Radwan, Ahmed G.

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Chaotic Clock Driven Cryptographic Chip: Towards a DPA Resistant AES Processor

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Approximation and realization of power-law all-pass filters

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Generalizing the Warburg impedance to a Warburg impedance matrix

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Optimum impedance spectroscopy circuit model identification using deep learning algorithms

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Extended Instantaneous Spectral Analysis (E-ISA) for Advanced Signal Processing

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

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Extended RC Impedance and Relaxation Models for Dissipative Electrochemical Capacitors

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Allagui, Anis; Benaoum, Hachemi; Elwakil, Ahmed S.; Alshabi, Mohammad

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2022

Synthesis of resonance-based common-gate fully differential band-pass filters

INTEGRATION-THE VLSI JOURNAL

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Fractional-order controllers for switching DC/DC converters using the K-factor method: Analysis and circuit realization

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Cengelci, Ekrem; Garip, Muhammet; Elwakil, Ahmed S.

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Estimating phase error using a Hilbert transform-based time-domain technique

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Implementation and Experimental Verification of Resistorless Fractional-Order Basic Filters

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2022

Versatile Field-Programmable Analog Array Realizations of Power-Law Filters

ELECTRONICS

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2022

Plant Tissue Modelling Using Power-Law Filters

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2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

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2022

FPAA-Based Realization of Filters with Fractional Laplace Operators of Different Orders

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Electrochemical stability analysis of red phosphorus-based anode for lithium-ion batteries

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Simple implementations of fractional-order driving-point impedances: Application to biological tissue models

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Ultra-low-power compact single-transistor all-pass filter with tunable delay capability

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Power law filters: A new class of fractional-order filters without a fractional-order Laplacian operator

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FPGA Realizations of Chaotic Epidemic and Disease Models Including Covid-19

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Quantification of memory in fractional-order capacitors

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On chip 0.5 V 2 GHz four-output quadrature-phase oscillator

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Employment of the Pade approximation for implementing fractional-order lead/lag compensators

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Enhanced FPGA realization of the fractional-order derivative and application to a variable-order chaotic system

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FPGA implementation of a chaotic oscillator with odd/even symmetry and its application

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Wideband third-order single-transistor all-pass filter

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Design and Implementation of an Optimized Artificial Human Eardrum Model

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Generalized Fully Adjustable Structure for Emulating Fractional-Order Capacitors and Inductors of Orders less than Two

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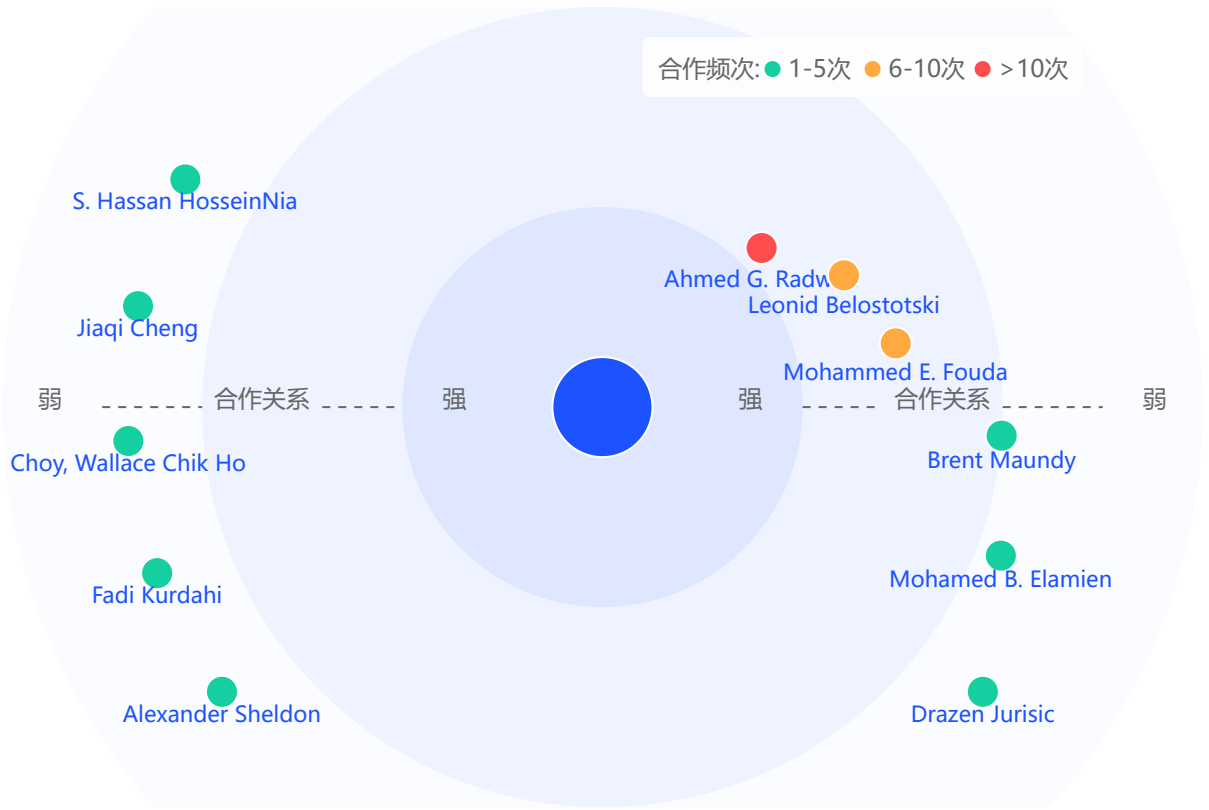
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